

Practical 3 - Parametric regression modelling

Survival Analysis

London School of Hygiene & Tropical Medicine

Datasets required: `whitehall`

R packages required: `survival`, `mexhaz`, `flexsurv`

```
library(survival)
library(mexhaz)
library(flexsurv)
```

Introduction

In this practical we will use the Whitehall data, which is familiar from the earlier practicals. We will investigate the association between job grade (`grade`) and risk of coronary heart disease (`chd`), with and without adjustment for age. We will use the time-in-study timescale except for in one question.

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>timebth</code>	Date of birth
<code>timein</code>	Date person entered the study
<code>timeout</code>	Date of the end of follow-up due to either death or censoring
<code>chd</code>	Event indicator at the end of follow-up: 0=alive or died from other cause, 1=death due to coronary heart disease
<code>grade</code>	Job grade at study entry. 1=professional/executive; 2=clerical/other
<code>agein</code>	Age in years at study entry

Load the data using:

```
whl<-read.table("datasets/whitehall.csv",sep=" ",header=T)
```

Aims

The aims of this session are:

- Fit exponential and Weibull distribution models to survival data and interpret the results.
- Use plots to assess whether a Weibull model is appropriate.
- Assess the constant hazard assumption of the exponential model.
- Understand the effect of changing the analysis timescale on estimates from the exponential model.

Question 1

We will begin with non-parametric analyses.

- Format the dates as in Practicals 1 and 2.
- How many individuals are there in each job grade group?
- Use a Kaplan-Meier plot to compare survival in the two groups. Interpret the plots. What is the estimated probability of survival beyond 10 years in each job grade group?
- Use the log rank test to compare the estimated survivor curves in the two job grades.

Question 2

We will now fit an exponential model to the Whitehall data using job grade as an explanatory variable.

- Write down the hazard model.
- Fit the exponential model using `mexhaz` using the code shown below.
- Identify and interpret the hazard ratio estimate.
- Identify and interpret the estimate of λ .

```
#Fitting exponential model using mexhaz
whl$time<-as.numeric(whl$timeout-whl$timein)/365.25
whl.exp<-mexhaz(Surv(time=time,event=chd)~as.factor(grade),
               base="pw.cst",data=whl,print.level = 0)
summary(whl.exp)
```

Question 3

- Using the estimates of λ and β in model output above, calculate (by hand) an estimate of the probability of survival beyond 10 years in the two job grade groups.
- Check your answer using the code below.
- If you have extra time at the end, have a look at the examples in the lecture notes to see other ways in which exponential models can be fitted in R, and how corresponding survival curves can be obtained.

```
#Obtain survival probabilities
surv.grade1.exp<-predict(whl.exp,
                        data.val = data.frame(grade=1),
                        time.pts = 0:19)

surv.grade2.exp<-predict(whl.exp,
                        data.val = data.frame(grade=2),
                        time.pts = 0:19)

#survival probability estimates at 10 years
surv.grade1.exp$results[10,]
```

	time.pts	as.factor(grade)		hazard	hazard.inf	hazard.sup	surv
10	10	1	0.004425163	0.00359919	0.005440688	0.9567132	
			surv.inf	surv.sup			
10	0.9470467	0.9646481					

```
surv.grade2.exp$results[10,]
```

	time.pts	as.factor(grade)		hazard	hazard.inf	hazard.sup	surv
10	10	2	0.008809281	0.006895095	0.01125487	0.9156759	
			surv.inf	surv.sup			
10	0.8935538	0.9333725					

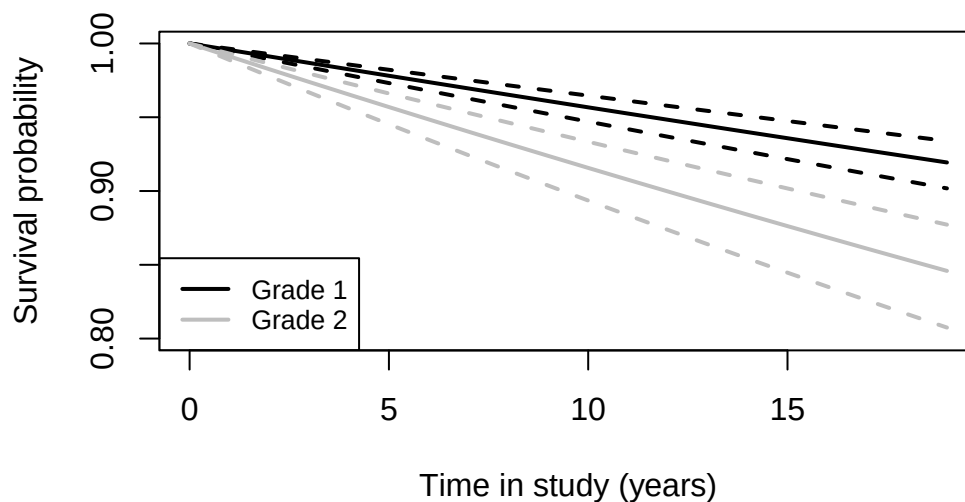
```
#Plot estimated survival curves
plot(0:19,c(1,surv.grade1.exp$results$surv),type="l",
     col="black",lwd=2,ylim=c(0.8,1),
     xlab="Time in study (years)",ylab="Survival probability")
points(0:19,c(1,surv.grade2.exp$results$surv),type="l",
       col="grey",lwd=2)
points(0:19,c(1,surv.grade1.exp$results$surv.inf),type="l",
       col="black",lwd=2,lty=2)#CI
```

```

points(0:19,c(1,surv.grade1.exp$results$surv.sup),type="l",
      col="black",lwd=2,lty=2)#CI
points(0:19,c(1,surv.grade2.exp$results$surv.inf),type="l",
      col="grey",lwd=2,lty=2)#CI
points(0:19,c(1,surv.grade2.exp$results$surv.sup),type="l",
      col="grey",lwd=2,lty=2)#CI

legend("bottomleft",
      c("Grade 1","Grade 2"),
      col=c("black","grey"),lwd=2,cex=0.8)

```



Question 4

Change to using age as the time scale (accounting for delayed entry into the study) and refit the exponential model. Compare your results with those found when using time-in-study as the timescale, and explain the results.

Question 5

Go back to the time-in-study timescale for this question and all subsequent questions. By fitting an exponential distribution we are assuming that the hazard rate does not change over time. Because this may not be a reasonable assumption we investigate fitting a Weibull model.

- Write down the form of the hazard model.
- Fit the Weibull model by using `base="weibull"` in `mexhaz`.

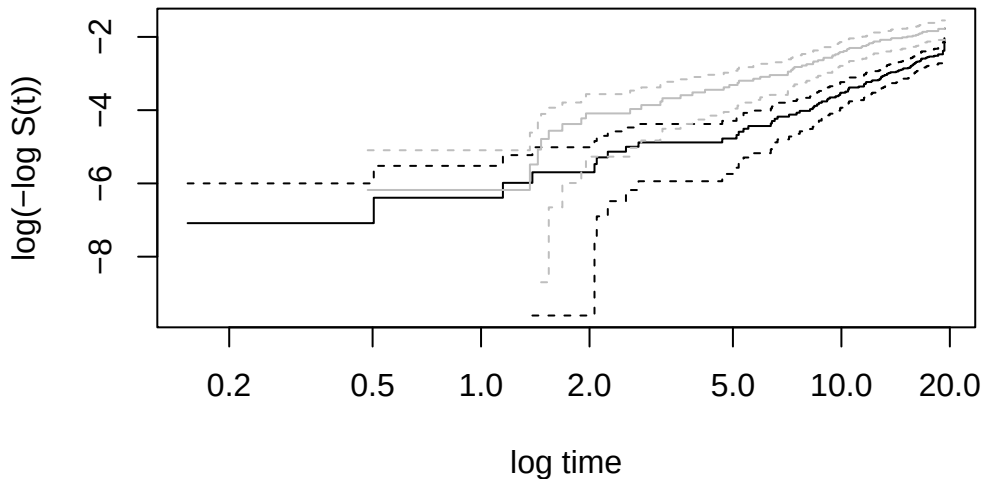
- What are the parameter estimates? Interpret the hazard ratio estimate.
- Use the results to compare the fit of the exponential and Weibull models.

Question 6

Use the code below to generate a plot that helps to investigate whether you expect the Weibull model fitted above to be appropriate.

- What is being plotted?
- Does the Weibull model appear to provide a good fit?

```
whl.km <- survfit(Surv(time=time,event=chd)~as.factor(grade),
                  data=whl)
plot(whl.km,conf.int=T,fun="cloglog",
     col=c("black","grey"),
     xlab="log time", ylab="log(-log S(t))")
```



Question 7

The age at which individuals entered the study may have an important part to play in explaining the time to event.

- Add age as an additional variable in the Weibull model.
- Interpret the hazard ratios for job grade and for age.
- What effect does adjusting for age at entry to the study have on the hazard ratio for job grade? Can you explain why this might happen?

- Based on this model, obtain estimated survivor curves for individuals in job grades 1 and 2, and aged 50 and 60. You can use the code below to obtain the estimates. Interpret the plots.

```
#Weibull model including grade and age
whl.wei.age <- mexhaz(Surv(time=time,event=chd)~as.factor(grade)+agein,
                     base="weibull",data=whl,print.level = 0)

#estimated survivor curves for individuals with
# job grade 1 and 2, aged 50 and 60.
#Shown for 1 individual - you can expand this for the others.
newdata<-expand.grid(grade=c(1,2),agein=c(50,60))
pred.wei.1<-predict(whl.wei.age,data.val=newdata[1,], time.pts=seq(0,19,1))
```

Question 8

The Weibull model allows the baseline hazard to change over time. A different approach is to split the follow up time into periods and fit an exponential model within each period, but allowing hazard ratios to be the same in each period. In other words, to allow the baseline hazard to differ in the different periods. This called a *piecewise exponential model*. It can then be investigated whether the baseline hazard changes across the periods. One way to do this is to create a record for each individual within each time period up to their event or censoring time. This approach of fitting exponential models within time bands is sometimes called *Lexis expansion*.

- Use the code below to set the data up for fitting a piecewise exponential model, and then to fit the model.
- Interpret the results. What is the baseline hazard in the four time periods? Notice that age has been centered at age 60 to help make the interpretation more sensible.
- Compare the results from this model with those from the Weibull model fitted earlier.

```
#Take a look at two example individuals
whl[whl$id %in% c(5001,5350),c("id","timein","timeout","time","chd")]

#Split the data into several rows per person
whl.split<- survSplit(Surv(time,chd)~., whl, cut=c(5,10,15,20),
                      start="tstart",end="time", episode="period")

#Take a look at what has happened to the data for the two example individuals
whl.split[whl.split$id %in% c(5001,5350),c("id","tstart","time","chd","period")]

#fit a piecewise exponential model, with age centered at age 60
```

```
whl.pw<-mexhaz(Surv(time=time,event=chd)~  
               as.factor(grade)+I(agein-60)+as.factor(period),  
               base="pw.cst",data=whl.split,print.level = 0)  
summary(whl.pw)
```